Speako : Digital Communications Platform

A Project Report Submitted

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DECLARATION

We hereby declare that the project work being presented in the project report entitled **“ Speako : Digital Communications Platform ”**.

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ABSTRACT

Communication skills play a pivotal role in personal, academic, and professional development. With the rapid adoption of digital platforms, there is a growing need for safe, user-friendly, real-time communication systems that empower individuals to practice speaking, interact confidently, and improve their conversational abilities. Existing platforms often pose limitations such as unsolicited messages, spam, privacy concerns, and lack of structured interaction models.

This project presents **SPEAKO**, a secure real-time communication platform built using the **MERN Stack (MongoDB, Express, React, Node)** integrated with **Stream Chat SDK** and **WebRTC-based video calling**. The platform introduces a **mutual connection model**, ensuring that communication happens only when both users accept a connection request, thereby improving safety and preventing harassment.

SPEAKO includes secure authentication, user profiles, friend request workflow, typing indicators, video calling, real-time messaging, and an admin module for monitoring user activity. This document provides detailed insights into system design, architecture, implementation, testing, and future enhancements.

This report aims to contribute to the development of modern, scalable, and secure communication systems while maintaining user trust, privacy, and a seamless user experience.

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INTRODUCTION

1.1 Background

The rapid advancement of internet technologies has transformed how individuals communicate, learn, and interact. Real-time communication platforms—video calling tools, chat applications, and social networks—have become integral to modern life. However, a significant gap exists in providing users with a *safe, structured environment to practice communication skills*.

Traditional social platforms focus on entertainment or casual messaging, lacking meaningful conversational guidance. On the other hand, tools like Zoom or Google Meet are built for formal meetings and are unsuitable for informal speaking practice. Language-learning apps lack human-to-human interaction, which is essential for improving fluency.

SPEAKO emerges as a solution that combines the reliability of modern web technologies with a *secure, permission-based communication model*.

1.2 Problem Statement

Despite the availability of chat applications and video conferencing tools, existing platforms have limitations:

1. **Unrestricted messaging** leads to spam, harassment, and unsolicited communication.
2. **Lack of a mutual-acceptance model** compromises user safety.
3. **Absence of practice-focused communication tools**, especially for individuals aiming to improve speaking skills.
4. **Limited integration of real-time chat + video calling** specifically designed for learning/conversational improvement.
5. **Privacy concerns and data misuse risks** in existing communication platforms.

There is a strong need for a platform that ensures safety, user control, simplicity, and real-time interaction while enabling users to build communication confidence.

1.3 Motivation

Human-to-human communication is the most natural and effective way to learn and improve conversational abilities. A safe environment is essential, especially for:

- Students practicing interviews
- Individuals learning new languages
- Professionals improving presentation skills
- People lacking social confidence

SPEAKO was motivated by the idea of creating:

- A trustable, secure, and controlled communication space
- A system where communication only occurs after mutual consent
- A real-time, interactive platform for voice + video + text practice
- A tool designed to improve communication skills safely

1.4 Objectives

3. To integrate real-time chat with typing indicators and read receipts.
4. To implement high-quality WebRTC-based video calling.
5. To provide easy profile management and user discovery features.
6. To ensure smooth communication using WebSockets and Stream APIs.
7. To offer an admin dashboard for system monitoring and user safety.

1.5 Scope of the Project

The scope of SPEAKO includes:

- Secure authentication using JWT
- User profile creation and editing
- Sending and receiving connection requests
- Text messaging
- Video calling between accepted users
- Administrative user management
- Real-time online presence indicators
- Notifications for all user events

Excluded from scope (future additions):

- Group calling
- AI-based communication analysis
- Mobile app version
- Gamification system

1.6 Expected Outcomes

Upon completion, SPEAKO will offer:

- A fully functional real-time chat + video based communication system
- A safe, moderated communication space
- Modern architecture using MERN + WebRTC + Stream SDK
- A scalable backend with clean API design
- A polished, responsive user interface
- High-quality academic + industrial-grade documentation

LITERATURE REVIEW

2.1 Overview of Real-Time Communication

Real-time communication enables instant transfer of data (audio, video, text) between users. Technologies include:

- WebSockets for bi-directional text communication
- WebRTC for peer-to-peer video/audio
- Real-time protocols for low latency

Applications: video conferencing, streaming, gaming, chat apps

2.3 Existing Communication Platforms

Platform	Strengths	Weaknesses
WhatsApp	Easy UI	Not designed for skill practice; random messaging
Zoom / Meet	Great for meetings	Too formal for casual speaking
Omegle	Random connections	Unsafe, lots of harassment
Duolingo	Language lessons	No real conversations

This highlights the need for a safe, controlled speaking practice platform.

2.4 Research Gap Identification

Existing solutions lack:

- Mutual-acceptance communication
- Skill-oriented features
- Real-time speaking practice tools
- Secure, structured conversation platform

SPEAKO fills this gap.

SYSTEM ANALYSIS

3.1 Requirements Analysis

Two categories:

Functional Requirements

- Registration
- Login
- JWT authentication
- Profile management
- Friend request workflow
- Real-time chat
- Video call
- Admin module

Non-Functional Requirements

- Security
- Low latency
- Scalability
- Usability
- Maintainability

3.2 Feasibility Study

1. **Technical Feasibility:** Uses MERN, Stream SDK, WebRTC — feasible with modern web technologies.
2. **Operational Feasibility:** Simple UI, responsive design, minimal user training required.
3. **Economic Feasibility:** Stream free plan sufficient; no heavy server resources needed.

SYSTEM IMPLEMENTATION

System implementation refers to the practical realization of the design specifications using appropriate technologies, tools, frameworks, and programming paradigms. SPEAKO is implemented using the **MERN Stack (MongoDB, Express, React, Node.js)** with real-time communication handled by **Stream SDK and WebRTC**.

This chapter provides a comprehensive breakdown of the implementation approach, module development, backend and frontend design patterns, API architecture, and real-time communication integration.

4.1 Overview of Implementation Strategy

The implementation of SPEAKO follows a modular and iterative development process:

1. **Backend Setup & API Development:**
   - Node.js + Express server initialization
   - JWT authentication flow
   - RESTful API development
   - MongoDB schema creation
2. **Frontend Setup:**
   - React app creation
   - Component architecture design
   - Tailwind CSS integration
   - Routing and protected routes
3. **Real-Time Messaging Integration:**
   - Stream Chat client setup
   - Channels and event listeners
4. **Video Calling Implementation:**
   - Stream Video SDK
   - WebRTC peer-to-peer negotiation
   - Call UI interface

This modular approach ensures scalability and reduces development complexity.

4.2 Backend Implementation (Node.js + Express)

The backend manages authentication, user management, connection workflow, and administrative operations. It follows RESTful principles and ensures all sensitive operations are secured using JWT middleware.

4.2.1 Project Structure (Backend)

```
/backend
|── controllers
|   ├── authController.js
|   ├── userController.js
|   ├── requestController.js
|   └── adminController.js
|── models
```

4.3 Database Implementation (MongoDB Atlas)

MongoDB is used for its flexible schema, scalability, and cloud availability.

4.3.1 User Schema

```
{
  name: String,
  email: String,
  password: String,
  profilePic: String,
  friends: [userId],
  createdAt: Date
}
```

4.3.2 Friend Request Schema

```
{
  senderId: ObjectId,
  receiverId: ObjectId,
  status: "pending" |
          "accepted" |
          "rejected",
  timestamp: Date
}
```

4.3.3 Admin Schema

Used for platform moderation.

4.4 Authentication Module

Authentication is implemented using **JWT (JSON Web Tokens)** stored in **HTTP-only cookies** for enhanced security.

Authentication Flow:

1. User logs in
2. Credentials validated
3. JWT created with user ID
4. Token returned to client
5. Protected routes check the token

Middleware functions:

- `verifyToken`
- `attachUserToRequest`

This ensures only authenticated users can access sensitive functionalities like chat, calls, and friend requests.

4.5 User Management

User management includes:

- Updating profile data
- Uploading profile pictures (Cloudinary)
- Fetching user information
- Searching users

4.5.1 Cloudinary Integration

Profile images are uploaded via Cloudinary:

Key operations:

- Upload image

4.6 Friend Request Workflow

The core safety feature of SPEAKO is the *mutual-acceptance communication model*.

Process:

1. User A sends a request
2. System checks if request already exists
3. Status = `Pending`
4. User B accepts → Chat + Call enabled
5. Reject → Request deleted

This workflow prevents unsolicited communication and enhances user privacy.

4.7 Chat Implementation (Stream Chat SDK)

SPEAKO uses Stream Chat SDK, which provides:

- 1-to-1 channels
- Real-time message updates
- Typing indicators
- Read receipts
- Online/offline presence

Chat Flow Implementation:

1. Initialize Stream client at frontend
2. Authenticate using Stream token
3. Create or open chat channel
4. Listen to:
   - `message.new` event
   - `typing.start` event
5. Render messages in real-time

Advantages:

- Zero WebSocket setup required
- Highly scalable infrastructure
- Developer-friendly API

4.8 Video Calling Implementation (Stream Video SDK + WebRTC)

Video calling is powered by WebRTC, wrapped through Stream's SDK for:

- Signaling
- Device handling
- Call-navigation
- UI rendering

4.8.1 How WebRTC Works in SPEAKO

Step 2: Signaling exchange

The SDK manages:

- ICE candidates
- Session descriptions (SDP)
- STUN/TURN server configuration

Step 3: P2P Connection

WebRTC establishes encrypted stream:

- Audio stream
- Video stream
- Bandwidth adaptation

Step 4: Ongoing Media Handling

Users can:

- Mute/unmute microphone
- Toggle camera
- End call

This setup ensures low latency and high-quality communication.

4.9 Frontend Implementation (React + Tailwind)

The frontend is built using component-based architecture for modularity.

4.9.1 Frontend Structure

```
/frontend
├── src
│   ├── components
│   ├── pages
│   ├── context
│   ├── services
│   ├── App.js
│   └── index.js
```

4.10 Routing & Protected Routes

Application pages include:

- `/login`
- `/register`
- `/dashboard`
- `/chat/:id`
- `/video/:id`

Protected routes check JWT validity before rendering components.

4.11 UI Component Implementation

- Notification popup
- Video call window

All UI components use:

- Tailwind for styling
- React Hooks for state
- Axios for API requests

4.12 API Design

SPEAKO provides REST APIs for all backend interactions.

Example Routes:

Route	Method	Description
`/api/auth/login`	POST	Login user
`/api/auth/register`	POST	Create new user
`/api/users/search`	GET	Search users
`/api/requests/send`	POST	Send connection request
`/api/requests/accept`	POST	Accept request
`/api/admin/users`	GET	Admin fetch users

APIs follow *REST conventions* and return JSON responses for easy integration.

4.13 Notifications Implementation

Notifications are implemented using Stream's channel events:

- New message
- New connection request
- Call received

Notifications update UI in real time using event listeners.

4.14 Error Handling & Logging

The system includes: Try/catch blocks, Mongoose error handlers, JWT error checks, HTTP status codes, and logging of server activity.

APPLICATIONS

SPEAKO, as a secure real-time communication and speaking-practice platform, offers a wide range of applications across various domains. Its controlled communication model, integration of live video calling, and safe environment make it suitable for individuals, educational institutions, corporate organizations, and training domains.

Applications:

- Improving spoken English or other languages
- Practicing pronunciation, accents, and fluency
- Real-time conversation with verified partners
- A safe environment for hesitant learners

5.2 Interview Preparation & Soft Skill Development

Many students and job seekers struggle with communication skills during interviews. SPEAKO can be used as an interactive mock interview environment where users can practice:

- HR round conversations
- Technical explanations
- Self-introduction
- Group discussion practice (future feature)

Because communication occurs only after mutual acceptance, the environment stays professional and secure.

5.3 Counseling & Mentorship Services

SPEAKO's secure chat + video call model can assist mentors, counselors, and trainers in interacting safely with learners or clients.

Potential use cases:

- Career counseling
- Academic mentorship
- Personal skill development sessions
- Online guidance for various subjects

Strict control of communication ensures privacy and user comfort.

5.4 Remote Collaboration

With the rise of remote work, SPEAKO can act as a lightweight communication tool for:

- Student project groups
- Remote teams
- Freelancers collaborating on tasks
- Peer-to-peer technical discussions

Unlike corporate tools, SPEAKO's simple interface makes it more intuitive for informal collaboration.

5.5 Social Confidence Building

Many individuals suffer from social anxiety or lack confidence in speaking. SPEAKO provides a *non-judgmental practice environment*, making it beneficial for:

- Individuals with stage fear
- People with limited social interaction
- Users improving casual communication skills

5.7 Professional Communication Training

Corporate organizations offer communication workshops for employees. SPEAKO can be used to:

- Practice presentation skills
- Conduct role-playing scenarios
- Improve cross-cultural communication
- Train customer support agents

Real-time interaction helps employees become more articulate and confident.

5.8 Mental Wellness and Peer Support (Future)

With proper moderation and AI monitoring, SPEAKO can be extended for:

- Peer support groups
- Emotional communication training
- Mental wellness interactions

The mutual-consent model ensures safe and respectful communication.

ADVANTAGES & LIMITATIONS

This chapter highlights both the strengths and limitations of SPEAKO. Understanding these helps evaluate the platform's current capabilities and identifies areas for future enhancement.

6.1 Advantages

1. Secure Communication Environment
SPEAKO uses JWT authentication and a mutual-acceptance model, ensuring privacy and preventing unwanted messages or calls.

2. Combines Chat + Video in One Platform
Users no longer need to shift across multiple apps. SPEAKO integrates:

- Real-time chat
- Voice/video calling
- Notifications

within a single platform.

3. Real-Time Performance
Stream SDK and WebRTC ensure: Low-latency calls, Instant delivery of messages, and Typing indicators and read receipts.

4. User-Friendly Interface
The UI is designed with simplicity and accessibility in mind, making it suitable for beginners and experienced users.

5. Scalable Architecture
MERN stack and cloud-based infrastructure allow easy expansion for thousands of users.

6. Safe for Learning & Practice
Beginners feel comfortable practicing communication skills because interactions occur only after mutual approval

The request-approval model blocks unsolicited contact, a major issue in traditional messaging platforms.

6.2 Limitations

1. One-to-One Communication Only: Currently limited to one-on-one conversations; no group calls or group chats.

2. Dependent on Internet Quality: Real-time video communication requires stable internet connectivity.

3. No AI Features Yet: SPEAKO does not include: AI feedback, Speech correction, or Automated conversation scoring. These may be added in the future.

4. Limited Administrative Tools: Admin moderation is functional but basic. Large-scale platforms require Auto-moderation and User behavior analysis.

5. No Mobile App: Currently web-only; mobile app development required for broader usability.

FUTURE SCOPE

Future scope outlines enhancements that can significantly improve SPEAKO's capabilities and platform reach. As technology evolves, the system can be expanded in various directions.

7.1 AI-Based Conversation Assistance

Integration of AI/ML techniques can provide: Speech-to-text analysis, Grammar and fluency evaluation, AI feedback on pronunciation, and Personalized suggestions. This would make SPEAKO a fully intelligent speaking companion.

7.2 Group Video Calling

Currently only one-to-one calls are supported. Future development includes:

- Group sessions
- Classroom-style virtual rooms
- Team communication spaces

7.3 Mobile Application (Android & iOS)

A cross-platform mobile app using **React Native** or **Flutter** will increase accessibility and user base.

7.4 Gamification Features

To increase engagement:

- Daily speaking challenges
- Communication badges
- XP points, Gamification motivates consistent practice.

7.5 Smart Matchmaking Algorithms

Pairing users based on:

- Skill level
- Interests
- Language preferences
- Personality type

This enhances compatibility and conversation quality.

7.6 End-to-End Encrypted Calls

Although WebRTC provides security, advanced encryption layers can be added for sensitive discussions and corporate users.

7.7 Admin Automation Tools

Machine learning can be used to detect: Harassment, Abuse, and Fake profiles. Automated moderation ensures platform purity.

CONCLUSION

SPEAKO is a modern, secure, and user-centered real-time communication platform designed to help users improve conversational skills while maintaining privacy and safety. The system integrates powerful technologies like **MERN Stack, Stream Chat, and WebRTC** to deliver high-quality messaging and video communication.

The unique **mutual-request approval system** distinguishes SPEAKO from conventional communication tools by eliminating unsolicited messages and promoting respectful interactions.

The platform successfully addresses key challenges in communication learning:

- Lack of safe practice spaces
- Anxiety during face-to-face communication
- Absence of integrated chat and video tools
- Privacy concerns in open messaging platforms

Through structured design, efficient implementation, and future expansion possibilities, SPEAKO proves to be a valuable solution for students, professionals, learners, trainers, and socially hesitant individuals.

With planned enhancements such as AI-based feedback, mobile applications, and group communication, SPEAKO has significant potential to evolve into a comprehensive global communication training ecosystem.

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